

Dimension in Commutative Algebra

Boran Erol

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Lemma 0.1. $\text{codim}(P) = \dim(R_P)$.

Proof. Prime ideals of R contained in P are in inclusion-respecting bijective correspondence with prime ideals of R_P , so we have the desired result. $\square$

Lemma 0.2. The dimension of R is the supremum of the dimensions of R_P where P is a maximal ideal of R .

In other words, dimension is a local concept.

Lemma 0.3 (Gathmann CA Lemma 11.6). Let R be a ring of finite dimension in which all maximal chains of prime ideals have the same length. Let P be a prime ideal of R .

Then, R/P is a ring of finite dimension where all maximal chains of prime ideals have the same length.

In particular, we have that $\dim(R) = \dim(R/P) + \dim(R_P)$.

Corollary 0.3.1. $\dim(R_P) = \dim(R)$ if and only if P is a maximal ideal.

Let's first show that integral ring extensions preserve the dimension of a ring.

Proposition 0.4 (Gathmann CA Proposition 11.9). All maximal chains of prime ideals in $K[x_1, \dots, x_n]$ have length n .

Proof. For $n = 0$, the statement is trivial. For $n = 1$, the statement follows from the fact that $K[x]$ is a PID. $\square$

Corollary 0.4.1. Let R be a finitely generated algebra over a field K . Assume R is an integral domain. Then, every maximal chain of prime ideals in R has length $\dim(R)$.

Proof. We can write R as $K[x_1, \dots, x_n]/P$ for some $n \geq 1$ and prime ideal P of R . $\square$

Let's now convert this statement to geometry.

Let Y be an irreducible subvariety of an irreducible variety X . Then, $\dim(X) = \dim(A(X)) = \dim(A(X)/I(Y)) + \text{codim}(I(Y)) = \dim(A(Y)) + \text{codim}_X(Y) = \dim(Y) + \text{codim}_X(Y)$.

Moreover, the local dimension of X is $\dim(X)$ at every point of X . In other words, all localizations of $A(X)$ at maximal ideals have dimension $\dim(X)$.

Definition 1. Let R be a ring. Let P be a prime ideal of R and $n \geq 1$. Let $P^{(n)} := \{a \in R : ab \in P^n \text{ for some } b \in R/P\}$. $P^{(n)}$ is called the n -th symbolic power of P .